

1 Main Ideas

Bootstrap Algorithm

1. At each main iteration t of the algorithm we should carry at each leaf ν :

- (a) A list $\mathcal{L}_t(\nu) = \{s_1, \dots, s_{N_{\nu,t}}\}$ of observed *optimization samples* at leaf ν .
 - i. Each s_i is obtained from randomly sampling the distributions **once** to obtain random sample ω_i (or getting the next sample from the list of samples we created originally) and solving the corresponding optimization problem:

$$\min_{x \in \mathcal{X}_{\nu}^{\mathcal{P}}} f(x, \omega_i), \quad (1)$$

where $\mathcal{X}_{\nu}^{\mathcal{P}}$ is the constraint partition set corresponding to leaf ν and $f(x, \omega_i)$ is the cost function of scheduling under crashing decision x and times given by ω_i .

- ii. In principle each leaf ν should carry its own list of samples $L_t(\nu)$ at every iteration t . We will initially test by having the list be the same for all samples and later allowing for different sample list among leafs. We will compare convergence of these two.

- (b) An expectation approximation

$$\begin{aligned} L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) &:= \frac{1}{N_{\nu,t}} \sum_{k=1}^{N_{\nu,t}} \min_{x \in \mathcal{X}^{\mathcal{P}}} f(x, \omega_k) \\ &= \frac{1}{N_{\nu,t}} \sum_{k=1}^{N_{\nu,t}} s_k \\ &= \frac{1}{N_{\nu,t}} \sum_{s_k \in \mathcal{L}_t(\nu)} s_k. \end{aligned} \quad (2)$$

At iteration $t + 1$ we will update $L(\mathcal{X}_{\nu,t}^{\mathcal{P}})$ by just adding to the sum the new optimization samples calculated in iteration $t + 1$ -step (a) and multiplying by $\frac{N_{\nu,t}}{N_{\nu,t+1}}$.

(c) A standard variance approximation $\text{Var}_t(\nu)$ with confidence interval

$$[L_{\nu,t}, U_{\nu,t}] = [L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) - \sigma q_{1-\alpha/2}, L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) + \sigma q_{\alpha/2}], \quad (3)$$

where σ is the bootstrap standard error and $q_{\alpha/2}$, $q_{1-\alpha/2}$ are the $\alpha/2$ and $1 - \alpha/2$ studentized t-bootstrap quantiles, respectively.

In the next section we will give details on how to calculate all the necessary components to obtain $\text{Var}_t(\nu)$ and $[L_{\nu,t}, U_{\nu,t}]$.

2 Details

Calculating $\text{Var}_t(\nu)$ and $[L_{\nu,t}, U_{\nu,t}]$

Suppose that the algorithm is at main iteration t and ν is a leaf of the tree at step t . Let $\mathcal{X}_{\nu}^{\mathcal{P}}$ be the constraint partition set corresponding to ν . Furthermore, assume that ν was also a leaf at iteration $t - 1$, so we have previous observed optimization samples $\mathcal{L}_{t-1}(\nu) = \{s_1, \dots, s_{N_{\nu,t-1}}\}$ and expectation approximation $L(\mathcal{X}_{\nu,t-1}^{\mathcal{P}})$. Then we do:

1. **Sampling & Optimization:** Let h be the number of extra samples that we will take in the iteration at ν ¹. Obtain new distribution random samples $\omega_1, \dots, \omega_h$ and calculate $s'_1, \dots, s'_h$, where

$$s'_i = \min_{x \in \mathcal{X}_{\nu}^{\mathcal{P}}} f(x, \omega_i), \quad i = 1, \dots, h.$$

2. **Obtaining $\mathcal{L}_t(\nu)$ & $N_{\nu,t}$:** Let

$$\mathcal{L}_t(\nu) = \mathcal{L}_{t-1}(\nu) \cup \{s'_1, \dots, s'_h\} = \{s_1, \dots, s_{N_{\nu,t}}\}, \quad (4)$$

where the 2nd equality is obtained by letting $N_{\nu,t} := N_{\nu,t-1} + h$ be the total number of samples and setting $s_{N_{\nu,t-1}+i} := s'_i$, for all $i = 1, \dots, h$.

¹ h is a parameter of the computation and we should be able to change this to make computational comparisons.

3. **Obtaining Sample Mean $L(\mathcal{X}_{\nu,t}^{\mathcal{P}})$:** Let

$$\begin{aligned} L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) &= \frac{N_{\nu,t-1}}{N_{\nu,t}} L(\mathcal{X}_{\nu,t-1}^{\mathcal{P}}) + \frac{1}{N_{\nu,t}} \sum_{i=1}^h s'_i \\ &= \frac{N_{\nu,t-1}}{N_{\nu,t}} L(\mathcal{X}_{\nu,t-1}^{\mathcal{P}}) + \frac{1}{N_{\nu,t}} \sum_{i=N_{\nu,t-1}+1}^{N_{\nu,t}} s_i. \end{aligned} \quad (5)$$

4. **Bootstrap Sampling 1:** Sample with replacement B samples $\tilde{\mathcal{L}}_t^1, \dots, \tilde{\mathcal{L}}_t^B$, each of size $N_{\nu,t}$ from $\mathcal{L}_t(\nu)$ ². This gives us the following B *resamples*:

$$\begin{aligned} \tilde{\mathcal{L}}_t^1 &= \{\tilde{s}_1^1, \dots, \tilde{s}_{N_{\nu,t}}^1\} \\ \tilde{\mathcal{L}}_t^2 &= \{\tilde{s}_1^2, \dots, \tilde{s}_{N_{\nu,t}}^2\} \\ &\vdots \\ \tilde{\mathcal{L}}_t^B &= \{\tilde{s}_1^B, \dots, \tilde{s}_{N_{\nu,t}}^B\}, \end{aligned} \quad (6)$$

where each $\{\tilde{s}_1^i, \dots, \tilde{s}_{N_{\nu,t}}^i\}$, $i = 1, \dots, B$, is sample taken **with replacements** from $\mathcal{L}_t(\nu)$.

5. **Bootstrap Statistics 1:**

(a) For each $i = 1, \dots, B$, calculate a *bootstrap sample mean* $\tilde{L}^i$ from its corresponding bootstrap sample $\tilde{\mathcal{L}}_t^i$:

$$\begin{aligned} \tilde{L}^i &:= \frac{1}{N_{\nu,t}} \sum_{\tilde{s}_k^i \in \tilde{\mathcal{L}}_t^i} \tilde{s}_k^i \\ &= \frac{1}{N_{\nu,t}} \sum_{k=1}^{N_{\nu,t}} \tilde{s}_k^i. \end{aligned} \quad (7)$$

(b) Obtain the mean of bootstrap statistics $\tilde{L}$:

$$\tilde{L} := \frac{1}{B} \sum_{i=1}^B \tilde{L}^i \quad (8)$$

(c) Obtain the standard error σ of the bootstrap sample means:

$$\sigma := \left[\frac{1}{B} \sum_{i=1}^B (\tilde{L}^i - \tilde{L})^2 \right]^{\frac{1}{2}} \quad (9)$$

²The number B is another parameter of our program and we should be able to change it to make computational comparisons.

6. **Bootstrap Sampling 2 (Studentized- t Sampling³):** For each $i = 1, \dots, B$, sample with replacements B_2 ⁴ samples $\tilde{\mathcal{L}}_t^{i,1}, \dots, \tilde{\mathcal{L}}_t^{i,B_2}$, each of size $N_{\nu,t}$ from its corresponding **bootstrap resample** $\tilde{\mathcal{L}}_t^i$. See Table 1 for more details.

Resamples from $\tilde{\mathcal{L}}_t^1$	Resamples from $\tilde{\mathcal{L}}_t^2$	$\dots$	Resamples from $\tilde{\mathcal{L}}_t^B$
$\tilde{\mathcal{L}}_t^{1,1} = \{\tilde{s}_1^{1,1}, \dots, \tilde{s}_{N_{\nu,t}}^{1,1}\}$	$\tilde{\mathcal{L}}_t^{2,1} = \{\tilde{s}_1^{2,1}, \dots, \tilde{s}_{N_{\nu,t}}^{2,1}\}$	$\dots$	$\tilde{\mathcal{L}}_t^{B,1} = \{\tilde{s}_1^{B,1}, \dots, \tilde{s}_{N_{\nu,t}}^{B,1}\}$
$\tilde{\mathcal{L}}_t^{1,2} = \{\tilde{s}_1^{1,2}, \dots, \tilde{s}_{N_{\nu,t}}^{1,2}\}$	$\tilde{\mathcal{L}}_t^{2,2} = \{\tilde{s}_1^{2,2}, \dots, \tilde{s}_{N_{\nu,t}}^{2,2}\}$	$\dots$	$\tilde{\mathcal{L}}_t^{B,2} = \{\tilde{s}_1^{B,2}, \dots, \tilde{s}_{N_{\nu,t}}^{B,2}\}$
$\vdots$	$\vdots$	$\dots$	$\vdots$
$\tilde{\mathcal{L}}_t^{1,B_2} = \{\tilde{s}_1^{1,B_2}, \dots, \tilde{s}_{N_{\nu,t}}^{1,B_2}\}$	$\tilde{\mathcal{L}}_t^{2,B_2} = \{\tilde{s}_1^{2,B_2}, \dots, \tilde{s}_{N_{\nu,t}}^{2,B_2}\}$	$\dots$	$\tilde{\mathcal{L}}_t^{B,B_2} = \{\tilde{s}_1^{B,B_2}, \dots, \tilde{s}_{N_{\nu,t}}^{B,B_2}\}$

Table 1: Resulting $B \times B_2$ *resample* table where each $\{\tilde{s}_1^{i,j}, \dots, \tilde{s}_{N_{\nu,t}}^{i,j}\}$, $i = 1, \dots, B$, $j = 1, \dots, B_2$, is a sample of size $N_{\nu,t}$ taken **with replacements** from $\tilde{\mathcal{L}}_t^i$.

7. Bootstrap Statistics 2:

- (a) For each studentized bootstrap sample $\tilde{\mathcal{L}}_t^{i,j} = \{\tilde{s}_1^{i,j}, \dots, \tilde{s}_{N_{\nu,t}}^{i,j}\}$, $i = 1, \dots, B$, $j = 1, \dots, B_2$, calculate a *2nd bootstrap sample mean*:

$$\begin{aligned} \tilde{L}^{i,j} &:= \frac{1}{N_{\nu,t}} \sum_{\tilde{s}_k^{i,j} \in \tilde{\mathcal{L}}_t^i} \tilde{s}_k^{i,j} \\ &= \frac{1}{N_{\nu,t}} \sum_{k=1}^{N_{\nu,t}} \tilde{s}_k^{i,j}. \end{aligned} \tag{10}$$

- (b) For each $i = 1, \dots, B$, obtain the mean of 2nd bootstrap statistics $\chi := \{\tilde{L}^{i,1}, \dots, \tilde{L}^{i,B_2}\}$:

$$\tilde{\tilde{L}}^i := \frac{1}{B_2} \sum_{j=1}^{B_2} \tilde{L}^{i,j} \tag{11}$$

- (c) For each $i = 1, \dots, B$, obtain the 2nd bootstrap statistic standard error of χ :

$$\sigma^i := \left[\frac{1}{B_2} \sum_{j=1}^{B_2} \left(\tilde{L}^{i,j} - \tilde{\tilde{L}}^i \right)^2 \right]^{\frac{1}{2}} \tag{12}$$

³Since we do not know the variance of our optimization sample process, we rely on a student- t version of bootstrap to obtain better approximations. This basically squares the effort that we need to do per iteration so we should be careful when implementing this part. Also, we will code the simple bootstrap version (without studentized computations) and will compare convergence.

⁴This is a parameter of our code and we should be able to change it so we can make numerical comparisons.

8. **Quantile Estimation & Confidence Interval:** Create the Z -transform random variable

$$Z := \left\{ \frac{\tilde{L}^i - L(\mathcal{X}_{\nu,t}^{\mathcal{P}})}{\sigma^i} \mid i = 1, \dots, B \right\}. \quad (13)$$

Then do:

- (a) Obtain the Z -transform mean:

$$E := \frac{1}{B} \sum_{i=1}^B \frac{\tilde{L}^i - L(\mathcal{X}_{\nu,t}^{\mathcal{P}})}{\sigma^i}. \quad (14)$$

- (b) Obtain the Z -transform standard deviation:

$$SD := \left[\frac{1}{B} \sum_{i=1}^B \left(\frac{\tilde{L}^i - L(\mathcal{X}_{\nu,t}^{\mathcal{P}})}{\sigma^i} - E \right)^2 \right]^{\frac{1}{2}} \quad (15)$$

- (c) Let $\alpha \in (0, 1)$ be our desired confidence level and set $q_{\frac{\alpha}{2}}$ and $q_{1-\frac{\alpha}{2}}$ to be the $\frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$ quantiles of Z , respectively. The desired confidence interval is given by:

$$[L_{\nu,t}, U_{\nu,t}] = [L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) - \sigma q_{1-\frac{\alpha}{2}}, L(\mathcal{X}_{\nu,t}^{\mathcal{P}}) + \sigma q_{\frac{\alpha}{2}}], \quad (16)$$

where $L(\mathcal{X}_{\nu,t}^{\mathcal{P}})$ is the sample mean given in (5) and σ is the bootstrap standard error given in (9).

3 Ranking & Selection

At this point we have for each node ν at time t we have $[L_{\nu,t}, U_{\nu,t}]$, E , SD .

1. For every ν calculate

$$d_{\nu,t} := d[(E, SD), (0, 1)] = [E^2 + (SD - 1)^2]^{\frac{1}{2}}.$$

2. Assign probabilities based on the distance:

$$P(\nu, t) = \frac{d_{\nu,t}}{\sum_{\varsigma \in \mathcal{N}} d_{\varsigma,t}} \quad (17)$$

3. Select with probability distribution $\mathcal{P}(\nu) := P(\nu, t)$ a number s of nodes that we will sample from.